$$
\begin{aligned}
p &= 1 + \frac{O + R}{gP_0}, \qquad q = 1 + \frac{R}{gQ_0}, \\
d &= p - v, \qquad \tau\dot{d} = -(p-d)^{1/\alpha - 1} d, \\
v &= p - d, \quad L = v^{1/\alpha - 1}, \quad f = \tau\dot{p} + Lp, \\
I_{\mathrm{Hb,NL}} &= \tau\dot{q} + Lq - \frac{fE(f)}{E_0}, \\
E(f) &= 1 - (1 - E_0)^{1/f}.
\end{aligned}
$$